Performance Optimization of FAS-Assisted U2G Communications Using Reinforcement Learning

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***Abstract*—Fluid antenna systems (FAS) are emerging as a promising technology for sixth-generation (6G) networks due to their flexible and reconfigurable radiation characteristics. In this paper, we investigate the performance optimization of FAS-assisted unmanned aerial vehicle (UAV)-to-ground (U2G) communications. Unlike conventional static antenna configurations, the proposed framework leverages the unique ability of FAS to dynamically activate and adjust antenna ports, enabling optimization in U2G channels. A reinforcement learning approach based on an enhanced proximal policy optimization (PPO) algorithm, termed Memetic PPO, is introduced to efficiently explore the discrete FAS configuration space by integrating a local search mechanism. Key performance metrics, including channel capacity and modeling accuracy, are derived and analyzed. Simulation results demonstrate that the number of active ports and port spacing significantly affect both metrics, and the proposed Memetic PPO algorithm achieves superior configurations by effectively balancing these factors. These findings provide valuable insights for the design and optimization of FAS in practical UAV communication systems.**

***Index Terms*—Fluid antenna system, unmanned aerial vehicle, channel capacity, reinforcement learning, proximal policy optimization.**

I. Introduction

Despite the remarkable achievements of fifth-generation (5G), it still faces some shortcomings, such as an over-reliance on base stations and high costs. In contrast, sixth-generation (6G), as the next-generation communication technology, aims to integrate satellites to build a globally covered network and features low costs, holding promising development prospects. The fluid antenna system (FAS) is regarded as one of the key technologies for 6G and has demonstrated vigorous development momentum. The core concept of FAS lies in the reconfigurability of antenna shape and position, with similar ideas traceable back to the directionally reconfigurable waveguide antenna proposed in 1960. It was not until recent years that the concept of FAS was proposed in [1] and [2]. Essentially, FAS refers to any controllable fluid, conductive, or dielectric structure that can dynamically alter its shape and position to reconfigure key parameters such as gain, polarization, radiation pattern, and operating frequency, thereby enhancing signal transmission efficiency, spectrum utilization, and coverage. Currently, there exists substantial academic research on FAS. [3] indicated that under identical conditions, FAS exhibits higher secrecy capacity, lower outage probability, and superior energy efficiency compared to conventional antennas. Furthermore, several effective hardware implementations have emerged. For instance, the PB-FAS proposed in [4] offers low power consumption, low cost, and easy integration, making it suitable for large-scale smart city deployment. [5] found that in communication scenarios where base stations are equipped with fluid antennas, channel capacity can be enhanced by optimizing the antenna positions at both the base station and the user ends. [6] pointed out that when users are equipped with FAS, the scalability of the network can be improved by alleviating the base station's channel state information burden. [7] comprehensively presented the design, channel modeling, and experiments of an ER-FAS, the results demonstrated that their proposed ER-FAS performs effectively in both the near-field and far-field, thereby enhancing the overall spectral efficiency of the wireless communication system. Recent years have witnessed the rapid development of artificial intelligence (AI), and the deep integration of communication and AI aligns with the vision of 6G. [8] first introduced Bayesian regression into FAS and proposed S-BAR, a general, high-precision, low-pilot-overhead channel estimation method for FAS with strong robustness and adaptability.

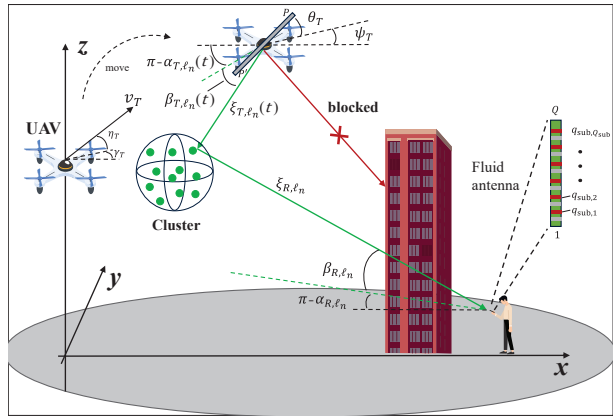

Fig. 1: Channel model of UAV-to-ground communication scenario assisted by FAS.

Although research on FAS is extensive, applying reinforcement learning (RL) to dynamic unmanned aerial vehicle (UAV)-to-ground (U2G) scenarios remains limited [8], [9]. To address the challenges of time-varying channels and the low exploration efficiency of baseline proximal policy optimization (PPO) in discrete action spaces, we propose an enhanced variant termed *Memetic PPO*. This algorithm

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integrates a local search mechanism to refine action selection before execution, significantly accelerating convergence and uncovering superior configurations. Furthermore, considering the model error discussed in [10], our framework optimizes FAS configurations by balancing the trade-off between channel capacity and modeling accuracy, providing better solutions for practical UAV communication systems.

Notation: The lowercase, boldface lowercase, and boldface uppercase letters such as x, $\mathbf{x}$, and $\mathbf{X}$ stand for the scalars, vectors, and matries, respectively. $\|\cdot\|$ represents the Frobenius norm, $\|\cdot\|_\infty$ represents the ℓ_∞ norm and $(\cdot)^{\mathrm{H}}$ denotes the conjugate transpose operation.

II. System Model

Consider a three-dimensional (3D) channel model for A2G communications as shown in Fig. 1, the UAV moves at a speed of v_T, and the azimuth and elevation angles of the direction of the motion are γ_T and η_T, respectively. In our model, the projection of midpoint of the UAV on the ground is defined as the coordinate origin, and the direction pointing to the receiver on the ground is the positive direction of the x-axis; the z-axis goes straight up and y-axis follows the right-hand rule. It is worth mentioning that the coordinate system does not move along with the UAV. Define H_0 and D_0 as the initial altitude of the UAV and the distance from the origin to the receiver, respectively.

On the UAV end, the azimuth and elevation orientation angles of the P omni-directional uniform linear array (ULA) antennas equipped on the UAV are respectively described as ψ_T and θ_T. The position of the p-th element in ULA can be expressed as

$$\mathbf{d}_p(t) = \mathbf{d}_T(t) + \mathbf{d}_{T,p}, \tag{1}$$

where $\mathbf{d}_T(t)$ denotes the time-varying position of the UAV, and $\mathbf{d}_{T,p}$ denotes the distance vector from the ULA midpoint to the p-th antenna. To be specific, they can respectively be expressed as

$$\mathbf{d}_T(t) = v_T t \begin{bmatrix} \cos\eta_T \cos\gamma_T \\ \cos\eta_T \sin\gamma_T \\ \sin\eta_T \end{bmatrix} + \mathbf{d}_{t_0}, \tag{2}$$

$$\mathbf{d}_{T,p} = k_p \delta_T \begin{bmatrix} \cos\theta_T \cos\psi_T \\ \cos\theta_T \sin\psi_T \\ \sin\theta_T \end{bmatrix}, \tag{3}$$

where $k_p = (P - 2p + 1)/2$, $\mathbf{d}_{t_0} = [0, 0, H_0]^{\mathrm{T}}$ and δ_T is the distance between adjacent antennas.

The receiver on the ground is equipped with a fluid antenna with a length of $W\lambda$ and Q ports, where λ is the carrier wavelength and $Q \gg Q_{\text{sub}}$, with Q_{sub} being the number of activated ports. The adjacent ports spacing of the fluid antenna Δ can be expressed as

$$\Delta = \frac{W\lambda}{Q-1}. \tag{4}$$

To characterize the dynamic port activation, we define a mapping relationship between the activated port index and the physical port index. Specifically, the q_{sub}-th activated port ($q_{\text{sub}} = 1, 2, \ldots, Q_{\text{sub}}$) corresponds to the q-th physical port ($q = 1, 2, \ldots, Q$) in the FAS through a mapping function $q = \mathcal{M}(q_{\text{sub}})$. Therefore, the position of the q_{sub}-th activated port relative to the global coordinate origin can be expressed as

$$\mathbf{d}_q = \mathbf{d}_R + k_q \Delta \begin{bmatrix} \cos\theta_R \cos\psi_R \\ \cos\theta_R \sin\psi_R \\ \sin\theta_R \end{bmatrix}, \tag{5}$$

where $k_q = (Q - 2\mathcal{M}(q_{\text{sub}}) + 1)/2$, ψ_R, θ_R are respectively the azimuth and elevation orientation angles of the FAS, and $\mathbf{d}_R = [D_0, 0, 0]^{\mathrm{T}}$ accounts for the position of receiver relative to the global coordinate origin.

The signal transmitted from the UAV will go through non-line of sight (NLoS) components, where the signal undergoes scattering by the cluster before arriving at the receiver. Due to the mobility of the UAV, the departure angles from the scatterers become time-varying. For a fixed scatterer position $\mathbf{d}_{\ell n} = [x_{\ell n}, y_{\ell n}, z_{\ell n}]^{\mathrm{T}}$ in the global coordinate system, the relative position vector from the UAV to the scatterer is given by

$$\mathbf{d}_{T,\ell n}(t) = \mathbf{d}_{\ell n} - \mathbf{d}_T(t), \tag{6}$$

the time-varying azimuth and elevation angles of departure (AoD) are then computed as

$$\alpha_{T,\ell n}(t) = \arctan2\left(y_{\ell n} - [\mathbf{d}_T(t)]_y,\ x_{\ell n} - [\mathbf{d}_T(t)]_x\right), \tag{7}$$

$$\beta_{T,\ell n}(t) = \arcsin\left(\frac{z_{\ell n} - [\mathbf{d}_T(t)]_z}{\xi_{T,\ell n}(t)}\right), \tag{8}$$

where $\arctan2(\cdot,\cdot)$ denotes the four-quadrant inverse tangent function, and $\xi_{T,\ell n}(t) = \|\mathbf{d}_{T,\ell n}(t)\|$ is the Euclidean distance between the UAV and the scatterer, given by

$$\xi_{T,\ell n}(t) = \sqrt{\sum_{i\in\{x,y,z\}} \left([\mathbf{d}_{\ell n}]_i - [\mathbf{d}_T(t)]_i\right)^2}, \tag{9}$$

here, $[\cdot]_i$ denotes the i-th component of the vector in the global coordinate system.

We can represent the channel matrix for the proposed channel model as $\mathbf{H}^{\text{cluster}}(t,\tau)$, where τ denotes the path delay. Each element $h_{p,q_{\text{sub}}}(t,\tau)$ in $\mathbf{H}^{\text{cluster}}(t,\tau)$ stands for the complex channel impulse response (CIR) of the propagation link from the p-th element in ULA to the q_{sub}-th element in FAS, which can be given as

$$h_{p,q_{\text{sub}}}(t,\tau) = h^{\text{cluster}}_{p,q_{\text{sub}}}(t)\delta\left(\tau - \tau^{\text{cluster}}(t)\right), \tag{10}$$

where $\tau^{\text{cluster}}(t)$ is the path delay for the NLoS propagation link, and $h^{\text{cluster}}_{p,q_{\text{sub}}}(t)$ can be expressed as

$$\begin{aligned} h^{\text{cluster}}_{p,q_{\text{sub}}}(t) = \sum_{\ell\in L}\sum_{n=1}^{\ell_N} & e^{j\varphi_{\ell_n} - j\frac{2\pi}{\lambda}(\xi_{T,\ell_n}(t)+\xi_{R,\ell_n})} \\ & \times e^{j\frac{2\pi}{\lambda}k_p\delta_T\cos(\alpha_{T,\ell_n}(t)-\psi_T)\cos\beta_{T,\ell_n}(t)\cos\theta_T} \\ & \times e^{j\frac{2\pi}{\lambda}k_p\delta_T\sin\beta_{T,\ell_n}(t)\sin\theta_T} \\ & \times e^{j\frac{2\pi}{\lambda}k_q\Delta\cos(\alpha_{R,\ell_n}-\psi_R)\cos\beta_{R,\ell_n}\cos\theta_R} \\ & \times e^{j\frac{2\pi}{\lambda}k_q\Delta\sin\beta_{R,\ell_n}\sin\theta_R} \\ & \times e^{j\frac{2\pi}{\lambda}v_T t\cos(\alpha_{T,\ell_n}(t)-\gamma_T)\cos\beta_{T,\ell_n}(t)\cos\eta_T} \\ & \times e^{j\frac{2\pi}{\lambda}v_T t\sin\beta_{T,\ell_n}(t)\sin\eta_T}, \end{aligned} \tag{11}$$

where $\{\varphi_{\ell n}\}_{n=1,2,\ldots,N}$ represents the random phase, assumed to be independent and uniformly distributed as $\varphi_{\ell n} \sim \mathrm{U}[-\pi, \pi)$. The terms $\xi_{R,\ell n} = \|\mathbf{d}_{\ell_n} - \mathbf{d}_R\|$ denote distances between the receiver and the n-th path in the ℓ-th cluster. Furthermore, α_{R,ℓ_n} and β_{R,ℓ_n} are angles of arrival. The scatterer positions follow the von Mises distribution

$$f(\alpha) = \frac{e^{\kappa \cos(\alpha - \mu_\alpha)}}{2\pi I_0(\kappa)}, \tag{12}$$

where κ is the concentration parameter, μ_α is the mean value of α, and $I_0(\cdot)$ is the modified Bessel function of order 0.

For U2G wireless communication systems, channel capacity is a crucial indicator of the achievable spectral efficiency, which is given by

$$C = \log_2 \left(\det \left(\mathbf{I}_{Q_{\text{sub}}} + \frac{\rho_{\text{SNR}}}{P} \overline{\mathbf{H}^{\text{cluster}}}(t,\tau) \overline{\mathbf{H}^{\text{cluster}}}^{\text{H}}(t,\tau) \right) \right), \tag{13}$$

where $\mathbf{I}_{Q_{\text{sub}}}$ denotes the identity matrix of size $Q_{\text{sub}} \times Q_{\text{sub}}$, ρ_{SNR} is the signal-to-noise ratio (SNR), and $\overline{\mathbf{H}^{\text{cluster}}}(t,\tau)$ represents the normalized channel matrix.

III. Memetic PPO Model

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